Younes SADDOUG

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21 years old

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4th year Engineering Student

(June\July internship)

EDUCATION

2022-October 2025	**Engineer in Applied Mathematics & Scientific Computing** *Sup Galilée Engineering School – Sorbonne Paris Nord University*
2023	**Bachelor's Degree in Mathematical Modeling and Engineering** *Institut Galilée – Sorbonne Paris Nord University*
2020-2022	**Preparatory Classes for Engineering Schools (CPGE) - Physics and Engineering Sciences** *Ibn Ghazi – Rabat, Morocco*
2020	**International Baccalaureate High School Equivalence : Physics Option**

PROFESSIONAL EXPERIENCES

Summer 2023
(2 months)

Al Barid Bank: Internship in the Business Intelligence Department – Predictive Analysis of Time Deposit Subscriptions

- Working with SQL for data manipulation and organization.
- Using sklearn.preprocessing for classification-focused preprocessing, addressing challenges like imbalanced data, scaling, and implementing Recursive Feature Elimination (RFE).
- Implementing of K-Nearest Neighbors and logistic regression algorithms for predictive modelling.
- Conducting a thorough comparative analysis using metrics and ROC curves to evaluate model performance.

PROJECTS

- **N-Body Problems - Astrophysics and Celestial Mechanics Simulation**
  - Exploring analytical solutions for specific cases of differential equations governing N-Body Problems.
  - Numerical simulation using the 4th order Runge-Kutta method and implementing differential equations in MATLAB.
  - Validating of simulations by comparing them to simple cases or analytical solutions.
  - Analyzing the accuracy and numerical stability of simulations.
  - Code optimization for resource-intensive cases.
  - Highlighting various planetary phenomena (e.g., gravitational resonance).

- **Participation in the 25th edition (2023/2024) of the Sup Galilée Enterprises Forum to present my major, Applied Mathematics.**

- **Solving the 1D Viscous Burgers' Equation through Deep Learning: Incorporation of Initial and Boundary Conditions.**
  - Understanding and applying the principles of supervised deep learning to physical problems.
  - Exploring deep learning methods for solving nonlinear PDEs (e.g., PINNs).
  - Developing a neural network model capable of predicting the dynamics of the viscous Burgers' equation for different boundary conditions.
  - Generating simulation data for training the model.
  - Evaluating the performance of the model and exploring improvements through fine-tuning.
  - Comparing the performance of DL with traditional numerical solutions of the equation (FDM,FEM,FVM).

SKILLS & TOOLS

Software

C / Python / R / Git / SQL / Matlab / Cuda
Power BI / Azure / AWS (ML & analytics services)

Technical skills

- ***Numerical analysis***
  - *Solving linear and non-linear systems using various methods.*
  - *Performance analysis (stability, convergence, error, benchmarking).*
  - *Parallel computing. (MPI , CUDA).*
- ***Partial differential equations***
  - *Advanced understanding of PDEs theory.*
  - *Application of numerical methods (FDM, FEM).*
- ***Stochastic processes , computing & Advanced understanding of probability theory***
- ***Optimization***
  - *Deep understanding of various algorithms for optimization problems.*
  - *Optimization methods in ML.*
- ***Statistics***
  - *Statistical computing.*
  - *Multivariate analysis.*
  - *Inferential statistics (e.g., Hypothesis Testing).*
  - *Statistical learning theory.*
- ***Data analysis***
  - *Extensive experience with Pandas, Numpy, Seaborn, Matplotlib libraries.*
- ***Machine learning \ Deep Learning \ AI***
  - *Design & implementation of different machine learning algorithms using Scikit-learn.*
  - *Experience with neural networks (TensorFlow & PyTorch experience).*
  - *Intermediate experience with NLP, computer vision.*
  - *Intermediate experience with HuggingFace library.*

Languages

English : *Advanced*, ***French*** : *Bilingual*, ***Arabic*** : *Bilingual*

HOBBIES & INTERESTS

High-Tech, Motorsports, Artificial Intelligence, Chess.